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## **Bachelor of Computer Applications**

**Semester – V**

**Business Analytics**

**Experiment No.: 3**

**Title: Building a Dashboard for depicting Visualisation and  
Deriving Insights**

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**GitHub Link:** <https://github.com/laishramritikumarsingh/BA-Assignment>

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## Experiment No.: 3

### Building a Dashboard for depicting Visualisation and Deriving Insights

#### Problem Statement

The objective of this experiment is to integrate business datasets from multiple sources into Microsoft Power BI and create an interactive sales analytics dashboard. The AdventureWorks datasets provide information about customers, products, sales transactions, product categories, product subcategories, and sales territories.

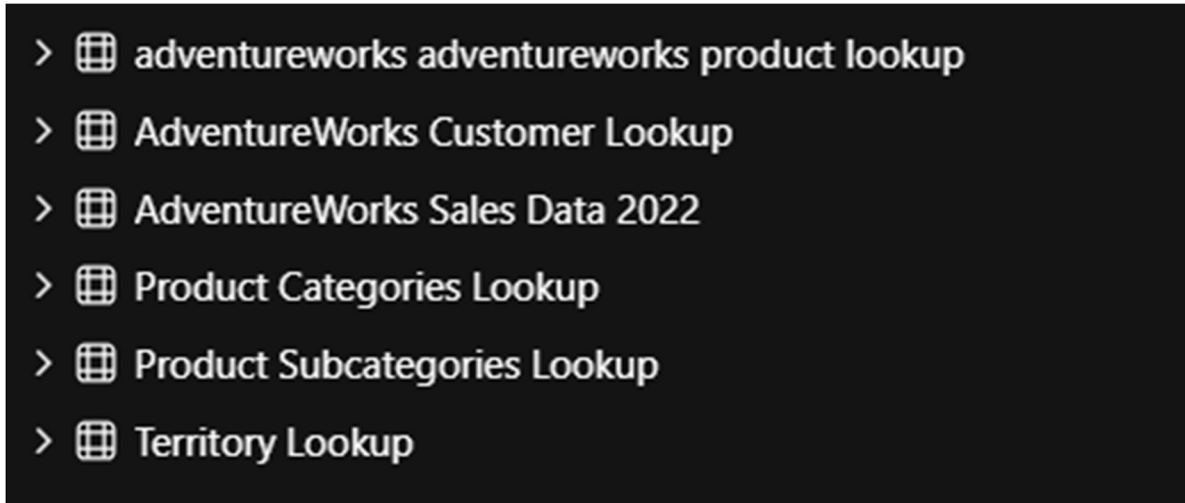
The datasets are connected to Power BI using MySQL and other available data sources. The collected data is then transformed, modeled, and analyzed to generate meaningful business insights into sales performance, customer behavior, product performance, categories, regions, and sales trends.

The experiment focuses on integrating multiple business data sources into Power BI to create a centralized reporting and analytics system.

#### Data Sources Used

| Dataset/Table                  | Purpose                                                  | Important Columns                                                |
|--------------------------------|----------------------------------------------------------|------------------------------------------------------------------|
| AdventureWorks Customer Lookup | Contains customer-related information                    | CustomerKey, FirstName, LastName, Gender, EducationLevel         |
| AdventureWorks Product Lookup  | Contains product-related information                     | ProductKey, ProductName, ProductColor, ProductCost, ProductPrice |
| AdventureWorks Sales Data 2022 | Contains sales transaction details                       | CustomerKey, ProductKey, OrderDate, OrderQuantity, TerritoryKey  |
| Product Subcategories Lookup   | Provides product classification at the subcategory level | ProductSubcategoryKey, ProductCategoryKey, SubcategoryName       |
| Product Categories Lookup      | Provides product classification at the category level    | ProductCategoryKey, CategoryName                                 |

| Dataset/Table    | Purpose                                        | Important Columns                             |
|------------------|------------------------------------------------|-----------------------------------------------|
| Territory Lookup | Contains geographical and regional information | SalesTerritoryKey, Region, Country, Continent |

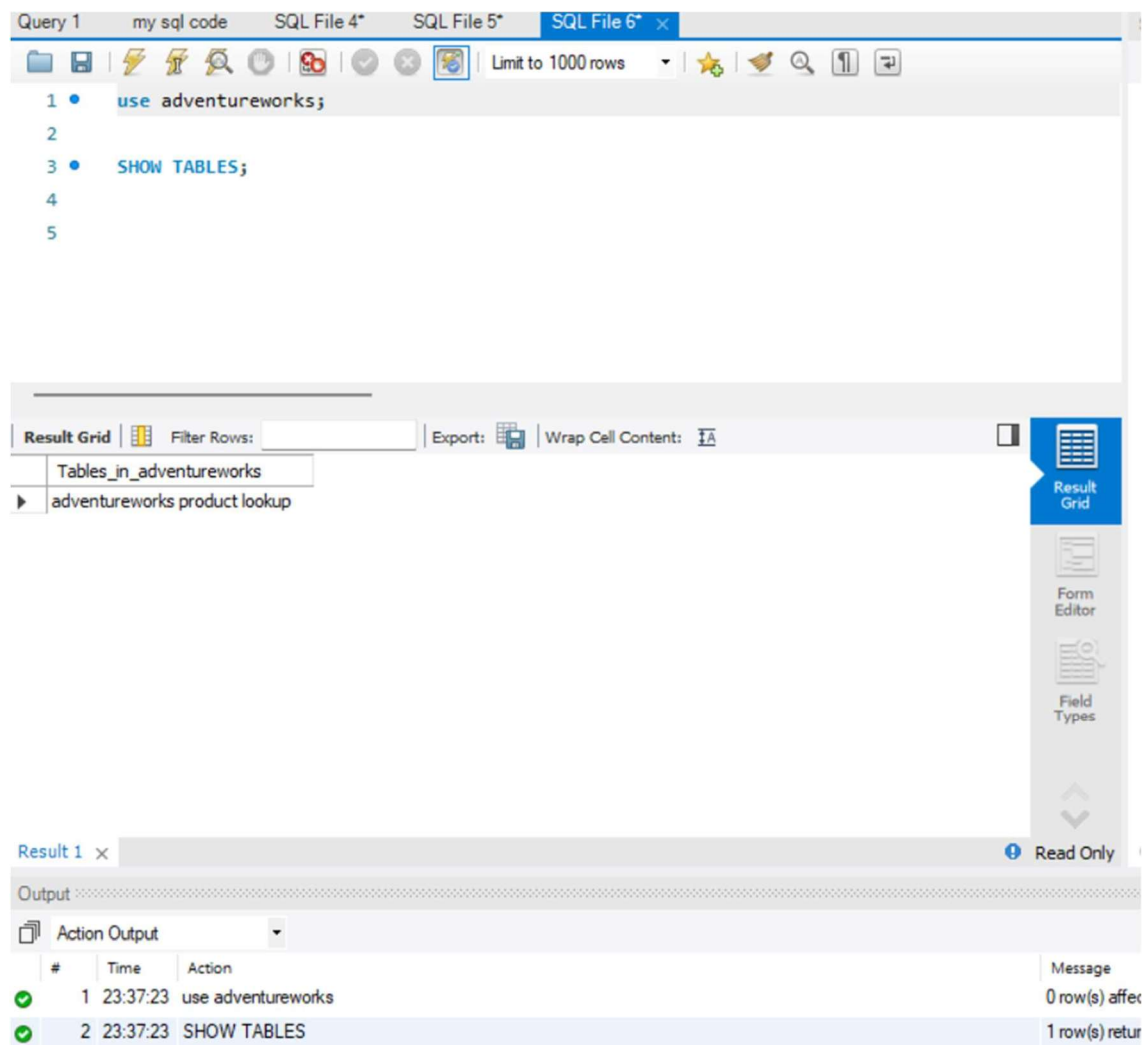


**Figure 1:** Data sources imported into Power BI

## Methodology

### ➔ Data Collection

- Collected AdventureWorks customer data.
- Collected product and product classification data.
- Collected sales transaction data for 2022.
- Collected sales territory information for regional analysis.
- The datasets were obtained from MySQL and other available data sources.



**Figure 2:** AdventureWorks tables available in MySQL Workbench

# Data Import

Power BI Desktop was opened and the required datasets were imported using the Get Data option. MySQL Database was selected as one of the data sources, and the required AdventureWorks tables were loaded into Power BI.

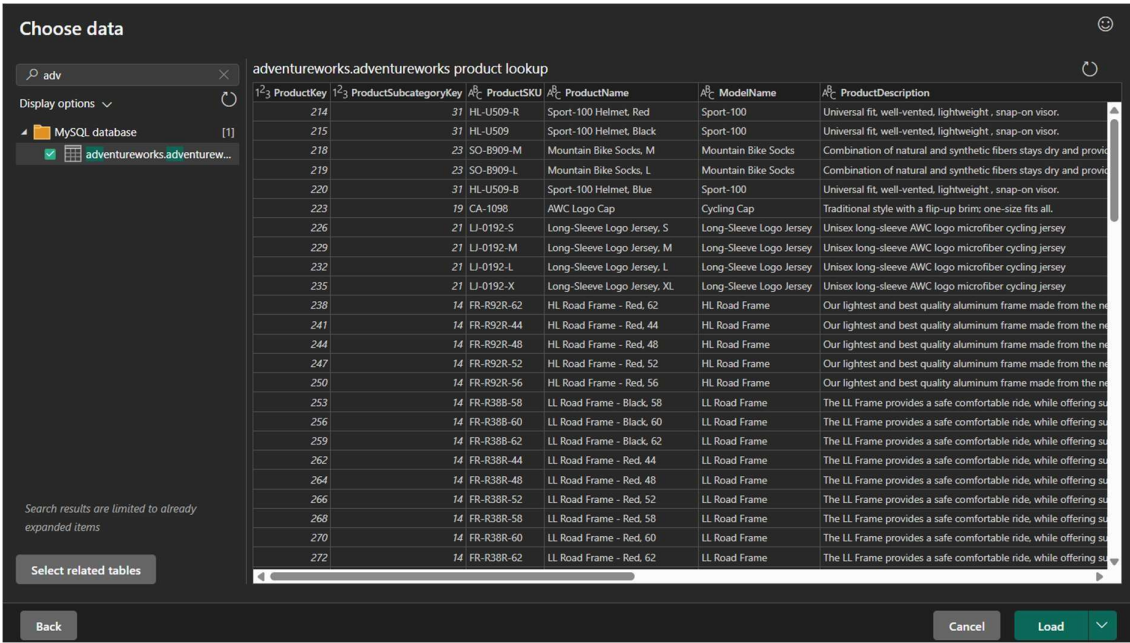
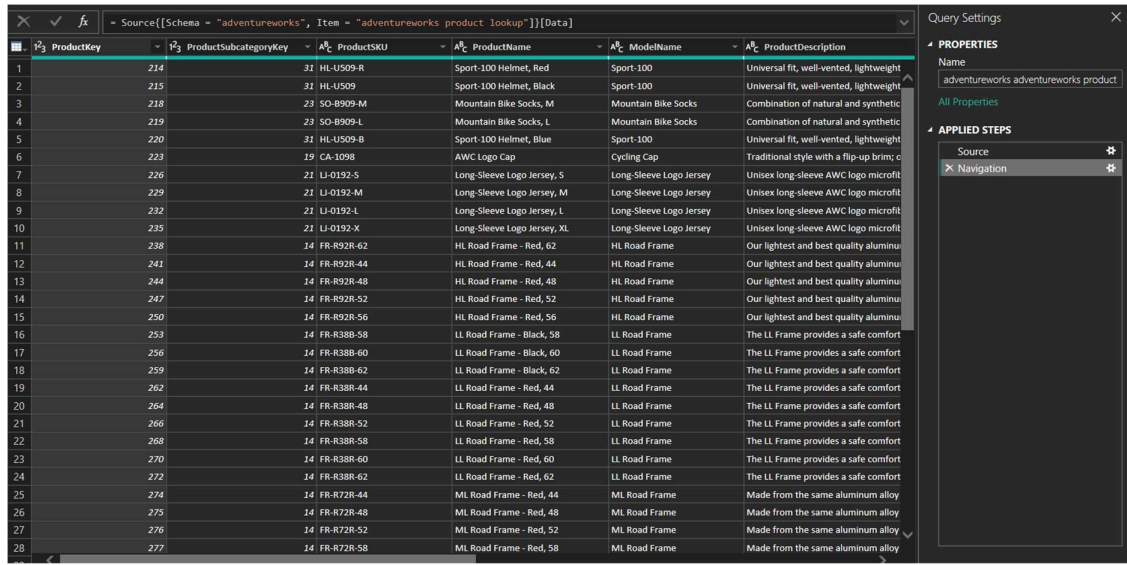


Figure 3: Importing AdventureWorks data from MySQL into Power BI

## ➔ Data Cleaning and Transformation

The imported datasets were reviewed and transformed using Power Query. Appropriate data types were assigned to numerical, text, and date columns. The data-quality indicators were checked to identify errors and empty values. Unnecessary rows were removed where required, and the datasets were prepared for further data modeling and analysis.



|    | ProductKey | ProductSubcategoryKey | ProductSKU | ProductName                 | ModelName               | ProductDescription                       |
|----|------------|-----------------------|------------|-----------------------------|-------------------------|------------------------------------------|
| 1  | 214        | 31                    | HL-U509-R  | Sport-100 Helmet, Red       | Sport-100               | Universal fit, well-vented, lightweight  |
| 2  | 215        | 31                    | HL-U509    | Sport-100 Helmet, Black     | Sport-100               | Universal fit, well-vented, lightweight  |
| 3  | 218        | 23                    | SO-B909-M  | Mountain Bike Socks, M      | Mountain Bike Socks     | Combination of natural and synthetic     |
| 4  | 219        | 23                    | SO-B909-L  | Mountain Bike Socks, L      | Mountain Bike Socks     | Combination of natural and synthetic     |
| 5  | 220        | 31                    | HL-U509-B  | Sport-100 Helmet, Blue      | Sport-100               | Universal fit, well-vented, lightweight  |
| 6  | 223        | 19                    | CA-1098    | AWC Logo Cap                | Cycling Cap             | Traditional style with a flip-up brim; c |
| 7  | 226        | 21                    | U-0192-S   | Long-Sleeve Logo Jersey, S  | Long-Sleeve Logo Jersey | Unisex long-sleeve AWC logo microf       |
| 8  | 229        | 21                    | U-0192-M   | Long-Sleeve Logo Jersey, M  | Long-Sleeve Logo Jersey | Unisex long-sleeve AWC logo microf       |
| 9  | 232        | 21                    | U-0192-L   | Long-Sleeve Logo Jersey, L  | Long-Sleeve Logo Jersey | Unisex long-sleeve AWC logo microf       |
| 10 | 235        | 21                    | U-0192-X   | Long-Sleeve Logo Jersey, XL | Long-Sleeve Logo Jersey | Unisex long-sleeve AWC logo microf       |
| 11 | 238        | 14                    | FR-R92R-62 | HL Road Frame - Red, 62     | HL Road Frame           | Our lightest and best quality aluminu    |
| 12 | 241        | 14                    | FR-R92R-44 | HL Road Frame - Red, 44     | HL Road Frame           | Our lightest and best quality aluminu    |
| 13 | 244        | 14                    | FR-R92R-48 | HL Road Frame - Red, 48     | HL Road Frame           | Our lightest and best quality aluminu    |
| 14 | 247        | 14                    | FR-R92R-52 | HL Road Frame - Red, 52     | HL Road Frame           | Our lightest and best quality aluminu    |
| 15 | 250        | 14                    | FR-R92R-56 | HL Road Frame - Red, 56     | HL Road Frame           | Our lightest and best quality aluminu    |
| 16 | 253        | 14                    | FR-R38B-58 | LL Road Frame - Black, 58   | LL Road Frame           | The LL Frame provides a safe comfort     |
| 17 | 256        | 14                    | FR-R38B-60 | LL Road Frame - Black, 60   | LL Road Frame           | The LL Frame provides a safe comfort     |
| 18 | 259        | 14                    | FR-R38B-62 | LL Road Frame - Black, 62   | LL Road Frame           | The LL Frame provides a safe comfort     |
| 19 | 262        | 14                    | FR-R38B-44 | LL Road Frame - Red, 44     | LL Road Frame           | The LL Frame provides a safe comfort     |
| 20 | 264        | 14                    | FR-R38B-48 | LL Road Frame - Red, 48     | LL Road Frame           | The LL Frame provides a safe comfort     |
| 21 | 266        | 14                    | FR-R38B-52 | LL Road Frame - Red, 52     | LL Road Frame           | The LL Frame provides a safe comfort     |
| 22 | 268        | 14                    | FR-R38B-58 | LL Road Frame - Red, 58     | LL Road Frame           | The LL Frame provides a safe comfort     |
| 23 | 270        | 14                    | FR-R38B-60 | LL Road Frame - Red, 60     | LL Road Frame           | The LL Frame provides a safe comfort     |
| 24 | 272        | 14                    | FR-R38B-62 | LL Road Frame - Red, 62     | LL Road Frame           | The LL Frame provides a safe comfort     |
| 25 | 274        | 14                    | FR-R72R-44 | ML Road Frame - Red, 44     | ML Road Frame           | Made from the same aluminum alloy        |
| 26 | 275        | 14                    | FR-R72R-48 | ML Road Frame - Red, 48     | ML Road Frame           | Made from the same aluminum alloy        |
| 27 | 276        | 14                    | FR-R72R-52 | ML Road Frame - Red, 52     | ML Road Frame           | Made from the same aluminum alloy        |
| 28 | 277        | 14                    | FR-R72R-58 | ML Road Frame - Red, 58     | ML Road Frame           | Made from the same aluminum alloy        |

Figure 4: product data transformation in Power Query

## ➔ Data Modeling

Relationships were created between the fact table and the corresponding lookup tables using common key columns. One-to-many relationships were used where the lookup table contains unique keys and the sales table contains multiple corresponding records.

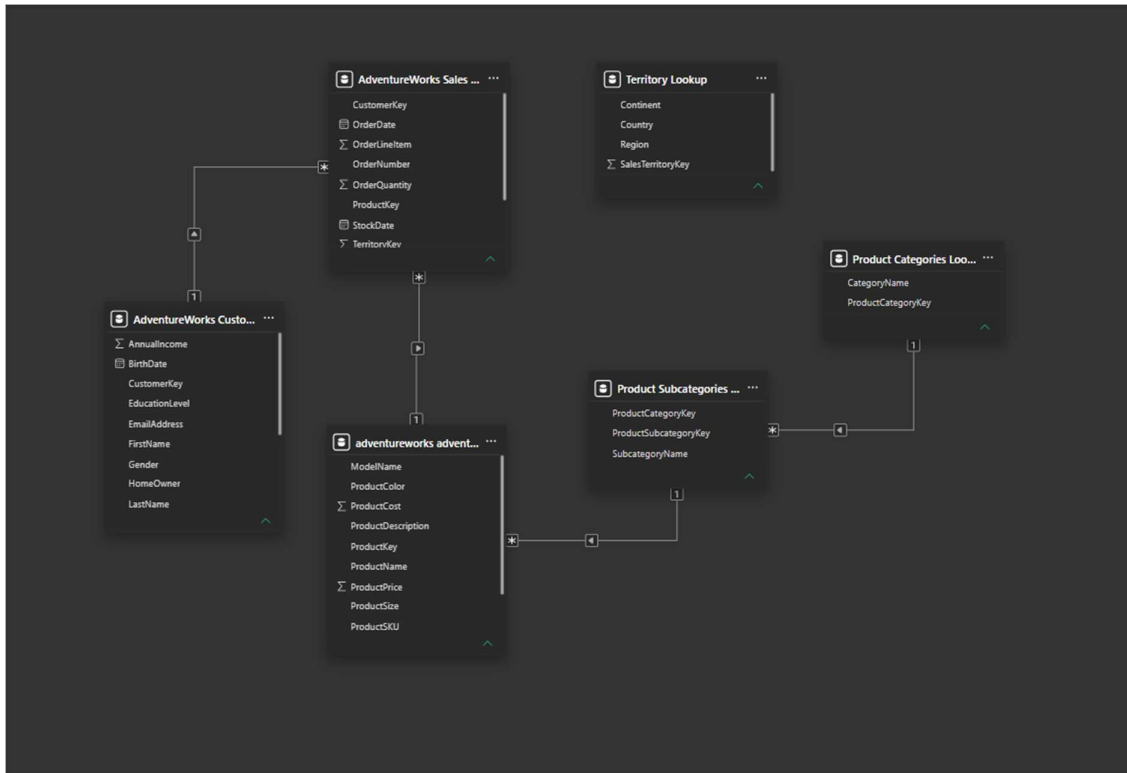


Figure 5: Data model and relationships in Power BI

## DAX Measures Used

### Total Sales

```
1 Total Sales =  
2 SUMX(  
3     'AdventureWorks Sales Data 2022',  
4     'AdventureWorks Sales Data 2022'[OrderQuantity] *  
5     RELATED('adventureworks adventureworks product lookup'[ProductPrice])  
6 )
```

## Total Orders

```
1 Total Orders =  
2 DISTINCTCOUNT('AdventureWorks Sales Data 2022'[OrderNumber])
```

## Total Customers

```
1 Total Customers =  
2 DISTINCTCOUNT('AdventureWorks Customer Lookup'[CustomerKey])
```

## Total Quantity Sold

```
1 Total Quantity Sold =  
2 SUM(  
3 | 'AdventureWorks Sales Data 2022'[OrderQuantity]  
4 )
```

## Average Order Value

```
1 Average Order Value =  
2 DIVIDE(  
3 | [Total Sales],  
4 | [Total Orders],  
5 | 0  
6 )
```

# Dashboard Output

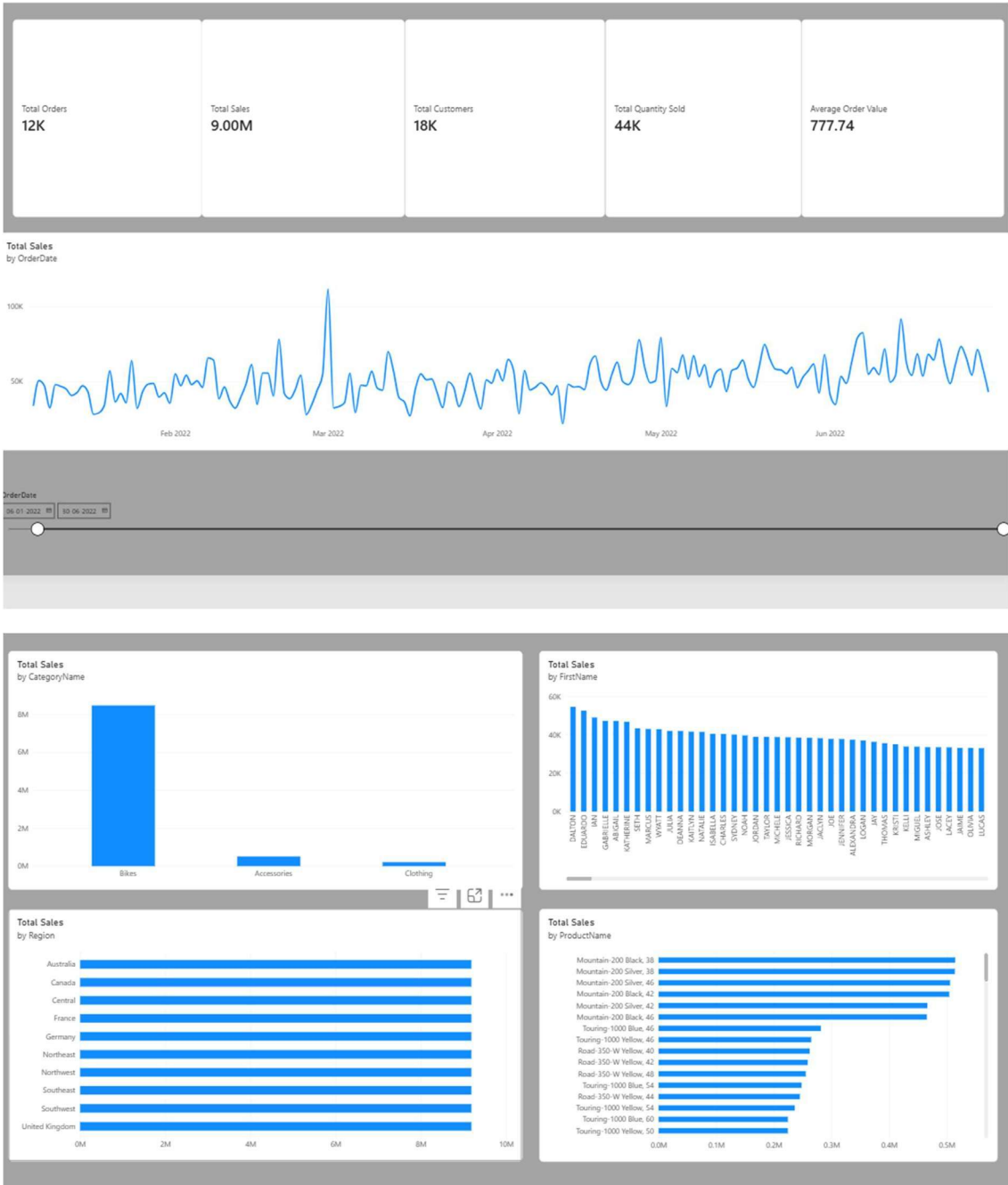


Figure 7: Final AdventureWorks Sales Dashboard

- **Total Sales Performance**  
The Total Sales KPI provides an overall view of the sales generated during the selected period.
- **Customer Analysis**  
Customer analysis helps identify customer-level sales performance and the contribution of individual customers to overall sales.
- **Product Analysis**  
The Top 10 Products by Sales visualization identifies products contributing the highest sales.
- **Regional Analysis**  
The Sales by Region visualization compares sales performance across different geographical regions.
- **Category Analysis**  
The Sales by Category visualization shows the contribution of different product categories to total sales.
- **Sales Trend**  
The Sales Trend Over Time visualization shows how sales vary across the selected dates.

## **Business Insights**

### **Key Business Insights:**

1. Total sales generated during the selected period were approximately ₹/currency 9.19M based on the dashboard.
2. The dashboard recorded approximately 12K orders.
3. Approximately 45K units were sold.
4. The Top 10 Products visualization highlights the products contributing most to sales.
5. Customer-level analysis identifies the highest-performing customers.
6. Sales trends vary across the 2022 period.
7. Regional and category analysis allows management to compare performance across different business segments.

## **Conclusion**

The experiment successfully demonstrated the integration of AdventureWorks business datasets into Microsoft Power BI using MySQL and other available data sources. The datasets were imported, transformed, and connected through appropriate relationships among the sales, customer, product, category, subcategory, and territory tables.

Various DAX measures were created to calculate important business KPIs, including Total Sales, Total Orders, Total Customers, Average Order Value, and Total Quantity Sold. An interactive dashboard was developed to analyze sales performance across different products, customers, categories, regions, and time periods.

The completed dashboard provides a centralized view of overall business performance and allows users to explore the data interactively using filters and slicers. This helps transform the integrated datasets into meaningful business insights and supports effective data-driven decision-making.

## **Learning Outcomes**

Through this experiment, I learned:

- How to import business datasets from MySQL and other data sources into Power BI.
- How to use Power Query for data cleaning and transformation.
- How to create relationships between multiple tables.
- How to establish one-to-many relationships in a Power BI data model.
- How to develop DAX measures for business KPIs.
- How to create interactive dashboards using Power BI visuals.
- How to use slicers and filters for interactive analysis.
- How to analyze sales performance by product, customer, category, region, and time.
- How to generate meaningful business insights from raw datasets.